
CREATIVELAB: UNIQUE CONCEPT GENERATION USING IMAGE CLUSTERING

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ABSTRACT

Image generation is a significant area of research with diverse applications in art, film, and education. Traditional methods for generating images often require manual intervention, particularly when adjusting for positive and negative constraints to enhance the quality and relevance of generated images. Richardson et al. (2023) in [1] propose an improvement by using Vision-Language Models (VLMs) under semi-supervised conditions to efficiently eliminate undesirable features. However, these methods still involve semi-supervised mechanisms that require some level of manual input. In contrast, our study introduces a novel approach that utilizes purely unsupervised learning methods. We employ k-means clustering to analyze and cluster image embeddings from various generated images. The cluster centers derived from this process are then used as negative prompts, against positive text prompts, to generate unique and relevant images. This method not only eliminates the need for semi-supervised learning but also significantly reduces the dependency on manual intervention, streamlining the image generation process and enhancing the consistency of the outputs. Our process additionally eliminates the need for BLIP2, as necessitated in [1]. As seen in our quantitative analysis, our method is completely unsupervised and does not need any specific prompting to cluster the generated images, while generating similarly creative imagery.

1 Introduction

The generation of images has tended towards the replication of pre-existing concepts due to its very nature. Training sets typically comprise natural images, thereby innately limiting the output to pre-existing concepts. By this limitation, the desire to develop models that can generate images that fall under conceptual norms, yet are totally imaginary in their existence, has increased over time, especially with the massive improvements in text-to-image generation. Consequently, work has been done to advance the generation of creative images and concepts that branch away from classical and standard expectations of a particular prompt [1]. These endeavors are grounded in the concept of computational creativity, which involves the replication, understanding, and modeling of creativity itself [2].

A wide variety of approaches have been taken to address the question of determining how best to model creative text-to-image generation. [3], for instance, proposed a "Creative Adversarial Network", in which they generated novel images that were not too distant from their distribution of original images, but would still increase stylistic ambiguity. This model operates like a GAN, but receives two signals, one to determine if some image is art or not and another to signal how well this image can be classified into pre-existing categories. More interestingly, however, [1] had determined such a method of generating unique images from concepts using diffusion priors and by continuously adding constraints to its optimization problem. Notably, their methodology, while effective, imposes significant time and power demands due to its frequent updates of the optimization problem. More prominently, these models necessitate a specific prompting to make advancements towards the final generated creative image. This is supervised, and due to this specific prompting, we can note that there is an assumption that a label must exist for some sub-concept. This assumption might not allow for any flexibility for non-creative images and sub-concepts that do not have a pre-defined label. Therefore, our objective is to develop an alternative methodology, building upon the groundwork laid by [1], which will ideally be more efficient in generating unique images from concepts, while not being tethered to labels, thus ideally being unsupervised.

In this paper, we seek to determine a quicker method of creative text-to-image generation built upon the pre-existing method by [1] that can bypass the lengthy process of continuously adding constraints to the optimization problem. In particular, we are interested in the model developed in [4], where a large set of images were generated of a particular prompt, which were then clustered together by their embeddings in the image domain. These clusters were used to determine cohesiveness, by which the authors could generate consistent characters. Interestingly to us, these clusters could also be considered to operate as "pre-existing sub-concepts" in our case of creative image generation. Thus, we could then generate a unique, imaginary sub-concept that is distant from these clusters in a latent space.

In speaking of a "concept", we are speaking of a generalized idea that can be subdivided into many sub-concepts. For instance, [1] uses the example of the concept "pet", where "pet" is the generalized concept, and then "dog" and "cat" could be considered sub-concepts which are pre-existing (i.e., not imaginary or particularly unique). We aim to develop a model that will generate a "pet" (concept) that is unique, meaning that it does not look like or take full attributes of a "pre-existing pet" (sub-concept).

In our case, our clustering based approach from [4] would imply that each cluster of images would refer to, at some level, a sub-concept. Going back to the pet example. One cluster of images could all be dogs or cats, alternatively we could have clusters that are even more specific, like white and black spotted dogs. Regardless, each of these clusters form pre-existing traits that we consider "negative" to our final image generation ("negative" in the sense that we do not want to generate images with similar features). In particular, the final image will be a "pet", but it will not be a "pet" that would fall into any of the found clusters, but would rather stand on its own. This, we imagine, would lead to imaginary and unique sub-concepts being generated.

2 Related Work

The motivation behind our concept for a model comes from a few key papers, but are built upon an immense prior of concepts and models for creative text-to-image generation, and text-to image generation more generally.

ConceptLab The primary model we are concerned about for development of our model is ConceptLab, which was developed in [1]. ConceptLab sought to perform creative text-to-image generation using a diffusion prior and an optimization problem that would be continuously modified. This model itself was built upon Kandinsky 2 [5], which itself uses a diffusion prior (in particular, the model consists of a text encoding, the use of an image prior to do an embedding mapping, and latent diffusion).

One particular concern with ConceptLab is that its diffusion prior constraints are generated over time. There are a set of positive constraints, C_{pos} , which contain a positive label (like "pet"), and a set of negative constraints, C_{neg} , which contain a set of pre-existing sub-categories (like "dog"). In this way, ConceptLab uses an evolutionary generation model,

where a set of images are generated for the concept itself, which then get encoded with CLIP, and then BLIP-2 VLM [6] is used to determine what type of sub-category it fits into within the broad category. Thus, if a new sub-category is identified (for example, "cat"), then that sub-category can be added to the negative constraints, C_{neg} . At each point, a new image is generated with the current C_{pos} and C_{neg} being used as the constraints for the diffusion prior, thus moving the image generated consistently towards originality.

The Chosen One: Consistent Characters in Text-to-Image Diffusion Models Another model that influenced our research was The Chosen One, as outlined in [4]. The purpose of this model was to generate consistent images of a certain character in various scenarios based on a text prompt. Specifically, once the model was given a text prompt, several images were generated using a text-to-image model that were later fed into a feature extractor. The embeddings from this feature extractor were then clustered using k-means, from which the most cohesive cluster was selected as the input for a personalization method. This personalization method was then used to generate a gallery of images for the next iteration. This process was repeated several times until convergence, effectively extracting a consistent identity that resonated with the original text prompt. This method allows for a specific character to be reliably generated across various settings, actions, and environments, and it notably outperforms other contemporaneous methods.

Primary applications of the model defined in [4] are producing illustrations for books, creating storyboards for a potential film, generating a consistent character in various poses, etc. Current limitations of this model include the inability to converge to consistent characters at random times and including unspecified or spurious attributes to a character. In our case, however, the primary consideration of this paper is the methodology by which it generates clusters of images. This procedure, by which it takes the embeddings of the images it generates and clusters them, will be employed by us to define different sub-concepts for the generalized concept we want to generate imaginary concepts for.

High-Resolution Image Synthesis with Latent Diffusion Models Latent diffusion models are able to generate high-quality images with lower computational demands by mapping high-resolution data to lower-dimensional latent spaces via an autoencoder [7]. The diffusion model aspect refers to a probabilistic model that learns some distribution by denoising some variable. The flexibility and performance of latent diffusion models make them a popular choice in other fields besides image generation, such as language modeling and video synthesis.

3 Methodology and Approach

The approach outlined in ConceptLab [1] utilizes positive and negative text-embedded features to compute the loss function, along with a new concept image-embedded feature to train the new concept text encoder. This process involves leveraging BLIP [6] to assign labels to images for the negative class using generated images, and utilizing the labeled text to generate text-embedded features as negative embeddings.

We aim to employ the methodology of clustering described in [4] to produce creative images more efficiently. In this way, a set of images describing the original concept itself are generated and then clustered by their similarities and shared features. This may, on face value, present an issue, because the negative embedded features generated through clustering may lack image labels. In reality, however, label information is unnecessary for negative embedding, as the objective is to remove all negative features from the new concept. By feeding all image features into the model, we can eliminate negative features within the image feature domain of the new concept. To do so, we can generate a cluster a set of images based upon their feature embeddings. We can then utilize the centroids of these clusters as the representative "feature set" of a particular sub-concept. This cluster centroid can then be used as the negative image features for our model. [4] clusters images, ensuring high similarity within each class in the image-embedded space, which can be utilized as our means of clustering images. Ideally, this approach allows us to designate which classes should serve as negative elements within the new concept. Figure 7 succinctly summarizes the workflow for our approach.

However, the question of maintaining the features necessary to still have the final creative image appear to be of the original class, we need a way to maintain that "concept" over time. Thus, we can employ a positive prompt. This positive prompt can be used by the model to leverage pre-existing image features to generate initial positive embeddings that can be utilized in the final concept. These feature embeddings ensure that the final image still exists within the desired original concept.

The original design from ConceptLab uses positive text embeddings and negative text embeddings to calculate cosine similarity with a new concept image embedding to update/learn the new concept text features. We don't learn the image embedding we wish to modify as might be expected, because the diffusion model will handle the image space and generate the image from the text feature by itself. If we update the new concept image embedding it also needs to

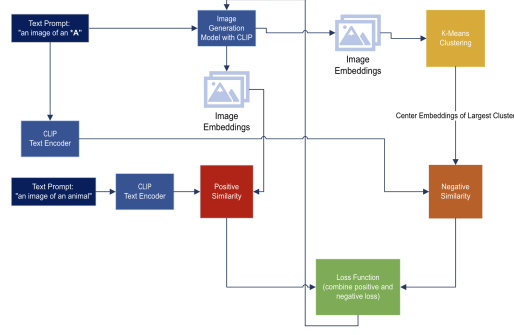


Figure 1: CreativeLab Workflow

fine-tune the diffusion model, which can take a lot of time compared to contrastive learning approaches. This is why ConceptLab only updates the new concept text features to achieve its goal.

We have restructured the pre-existing loss function and embedding spaces to prioritize image features over the original text features, aligning with recent advancements in feature extraction techniques. While positive embeddings continue to reside within the text domain to capitalize on their comprehensive feature coverage, we now extract negative embeddings directly from clustered image features, diverging from the traditional text-based approach as previously employed by BLIP2. Our embedding framework is rooted in the CLIP architecture [8], which ensures a robust correspondence between text and image modalities when representing identical concepts. Consequently, we have refined our embedding strategies for new concepts by utilizing image-based features for positive embeddings and text-based features for negative embeddings. Although this strategic adjustment introduces an increase in complexity and extends the processing time, it eliminates the need for BLIP2 within our model. This removal significantly accelerates the training process, optimizing overall efficiency.

3.1 Objective Function

The formulation of the loss function, as shown in Equation 1, is pivotal for this process. The total loss $\mathcal{L}$ is a combination of negative and positive losses, weighted by a factor λ :

$$\mathcal{L} = \mathcal{L}_{neg} + \lambda \cdot \mathcal{L}_{pos} \quad (1)$$

The Positive Loss $\mathcal{L}_{pos}$, detailed in Equation 2, is calculated by subtracting the cosine similarity between the new concept’s image embedding and the positive prompt’s text embedding from 1. This formulation ensures that the loss function is structured such that a lower value is indicative of a better outcome. By using $1 - \text{cosine similarity}$, we aim to minimize the loss, thereby maximizing the alignment of the generated image with the desired concept. This transformation of the similarity into a loss metric is critical for directing the model to enhance the coherence between the image and its corresponding textual description.

$$\mathcal{L}_{pos} = 1 - \text{cos}_{sim}(\text{image embedding}_{new}, \text{text embedding}_{pos}) \quad (2)$$

In this configuration, a lower $\mathcal{L}_{pos}$ indicates that the generated image closely matches the intended concept, as represented by the positive prompt. This approach effectively guides the model in generating images that are not only creative but also accurately reflect the characteristics embedded in the text prompt.

Conversely, the Negative Loss $\mathcal{L}_{neg}$, as specified in Equation 3, integrates both the mean and the maximum cosine similarities between the new concept’s text embedding and multiple negative image embeddings. This combined approach is designed to counteract the potential dilution of value that might arise when solely the mean similarity is used. This could arise in a scenario involving a large array of negative examples. Our methodology for this equation is influenced by ConceptLab [1], which emphasizes the influence of the class that exhibits the highest similarity. This is achieved by adding the maximum similarity to the calculated mean, enhancing the impact of the most similar, yet undesirable, concepts on the overall loss calculation. Such a strategy is pivotal in penalizing any resemblance to the closest undesired concepts, thereby aiding in the effective removal of such features from the new concept.

$$\begin{aligned}\mathcal{L}_{neg} = & (\text{mean } \cos_{sim}(\text{text embedding}_{new}, \text{image embeddings}_{neg}) \\ & + \max \cos_{sim}(\text{text embedding}_{new}, \text{image embeddings}_{neg})) \\ & * 0.5\end{aligned}\tag{3}$$

This approach of incorporating both the average and maximum similarities ensures that the model actively avoids alignment with any unrelated or undesirable concepts, thereby maintaining a higher level of purity in the differentiation process of the new concept from existing ones.

These metrics are critical for fine-tuning the model to minimize irrelevant similarities, while maximizing relevance to the conceptual descriptions, ensuring the images' uniqueness and cohesion.

3.2 Image Data Generation

In order to get a set of data that we can use to determine pre-existing categories to use as negative prompts, we can consider generating a set of images for the concept itself. We used a set of prompts, like "a photo of a {token}", where {token} refers to the name of the concept itself ("pet", for instance).

In order to test the image generation itself, We had chosen to generate a total of 2,500 images, using some set prompt, through primarily Stable Diffusion [7]. We had, however, additionally generated images via the Realistic Vision and Kandinsky 2 models using the same positive prompt. Within each of these generations, we had additionally tested image generation with negative prompts like "person," "human," "owner," and "people" (in the case where the token was "pet"), to generate images of just pets. Similar to the images generated via stable diffusion, the images from Realistic Vision contained primarily cats and dogs but in a more lifelike style [9]. Kandinsky 2 [5] generated primarily cats and dogs, but we had noticed that the images generated therein were fairly stylistically similar and were not as unique as those from the other two models.

With respect to our model itself, we will be generating some number of images that we have decided would be effective for a clustering. These images would be generated using Kandinsky 2, and only the clip embeddings of those images will be used for the clustering.

Something particular to note with our use case is that the number of images generated for each iteration is a parameter of consideration. The optimal parameter may very well differ from concept to concept, but in our particular case, we found that a set of 10 images being generated at each interval would be sufficient for our concepts. Thus, at a set of predefined intervals within the process of the model, a set of 10 images would be generated that would be used for the image clustering and thus as negative image features towards the final product. For this image generation, we had settled upon using Kandinsky 2, primarily because it was being used already in the original ConceptLab [1], but also because it was well defined with the clustering methodology we will outline in the next section.

3.3 Image Clustering

As previously explained, our methodology for image clustering is adapted from [4]. This is used to cluster a set of generated images that will define each of our positive prompts.

The data that we tested upon was a set of 2,500 generated images of the generalized concept, through the various different methods outlines in section 3.2. We tested, additionally, a comparison of the clustering of these images using either the embeddings of each of these images from dinov2 [10] or from CLIP.

These embeddings from dinov2 and CLIP are clustered using a k-means++ model, where the number of clusters are pre-determined by us. This determined set of clusters is a parameter that can be updated per concept that one may wish to generate for. For the particular set of concepts that we have chosen to generate images for, we have found that 3 clusters were efficient, after having attempted clustering with a wide variety of values for k . Generally, at very large numbers for k , we found that clusters tended to be separate on features that were not extremely relevant (for instance, we might have 10 different clusters for "dog" if the concept is "pet", where each cluster has different dog breed or color pattern). We found that limiting the number of clusters would instead allow us to focus on the overall generalized sub-concepts as opposed to each individual and particular (in a very minute sense) sub-concept that may exist.

For instance, we had clustered a set of 2,500 images of a "pet" generated by Kandinsky 2 ([7]), in particular using the prompt "an image of a pet". This image clustering used the CLIP embeddings. The clustering for this particular case can be seen in Figure 2.

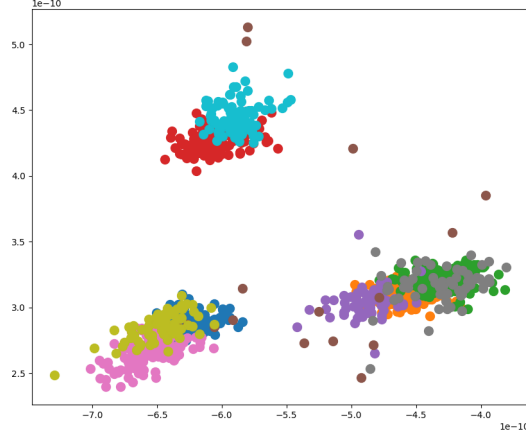


Figure 2: Kandinsky 2 Clustering for "an image of a pet" with CLIP

Once we have these clusters established, we can take the cluster centers, which themselves are an averaged image embedding that we consider to represent each of the images themselves that exist within the cluster. These cluster center embeddings can then be used as our negative prompts, representing the features of pre-existing items in the category.

With respect to the two different embedding methodologies (using dinov2 and CLIP), although dinov2 was what was initially used in the model outlined in The Chosen One [4], we found that, when clustering images generated by each of Stable Diffusion, Kandinsky 2, and Realistic Vision and while calculating embeddings across dinov2 and CLIP for each of those generation models, qualitatively, using CLIP embeddings for Kandinsky 2 images produced the most distinct and unique clusters. This suggests, in our purview, that the cluster centroids we take from this combination of image generation with embedding calculation would provide the most relevant and distinct features we may wish to generate away from on future iterations.

Additionally, the code used by [1] that we had built upon already uses CLIP embeddings throughout the original mode, so continuing with this methodology would be the simplest to integrate with.

4 Experiments and Results

In this section, we present our experimental design, results, and further evaluations that build upon the core CreativeLab framework. The experiments were structured to assess three key aspects: effectiveness of clustering-based negative prompt generation, sensitivity to hyperparameter tuning, and generalizability across different concept domains.

4.1 Training Dynamics and Hyperparameter Exploration

To evaluate training performance, we tested multiple configurations involving learning rates, lambda values, total steps, and cluster update intervals. Our base model is trained with 2,500 steps, following the original ConceptLab configuration. We experimented with varying the number of generated images and k values for clustering to determine their influence on output creativity.

Additionally, we examined the impact of varying the learning rate on creative image generation. Figure 3 and Figure 4 illustrate visual comparisons of images generated at different step counts with learning rates of 0.0001 and 0.01, respectively. These images show that higher learning rates sometimes lead to faster convergence but can produce more artifacts or inconsistent styles. Conversely, lower learning rates preserve stylistic coherence but may require more steps to achieve uniqueness.

4.2 Cluster Embedding and Prompt Influence Study

We conducted additional studies to compare embeddings from CLIP and DINOv2, and determined that CLIP yielded more distinguishable cluster centroids on images generated via Kandinsky 2. Figure 2 shows representative clustering output using CLIP.



Figure 3: Learning Rate = 0.0001 and Lambda = 0.8

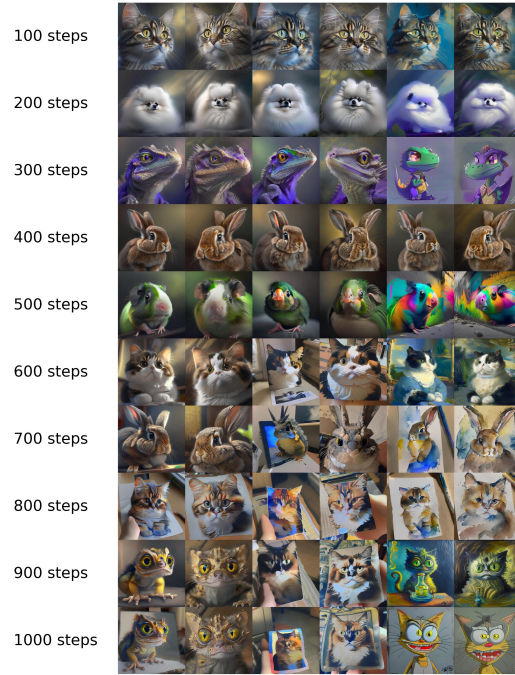


Figure 4: Learning Rate = 0.01 and Lambda = 0.8

We also tested the role of cluster centroids as negative prompts. When comparing models with and without centroid-based negative prompts, results indicated that centroid usage improves concept distinctiveness and reduces feature overlap with common sub-concepts like typical dog/cat features under the "pet" prompt.

4.3 Cross-Model Clustering Evaluation

We extended our evaluation to compare clustering quality from images generated by Stable Diffusion, Realistic Vision, and Kandinsky 2 using the same “an image of a pet” prompt. Figure 5 presents the results. Visually, clustering on Kandinsky 2 output provided clearer separation of sub-concepts, reinforcing its use in CreativeLab.

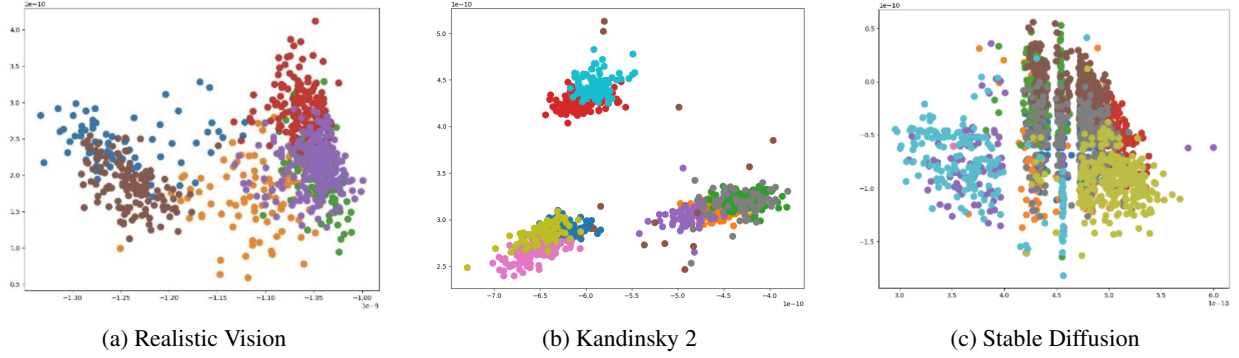


Figure 5: Clustering Results for "an image of a pet" Prompt

4.4 Human Evaluation and Prompt Diversity

Figure 6 demonstrates that our model generalizes beyond animal concepts to generate unique instances of buildings, fruits, and vehicles. We also plot CLIP-based similarity scores (Figure 7) between generated images and their textual prompts, showing that CreativeLab maintains conceptual relevance despite increased creativity.

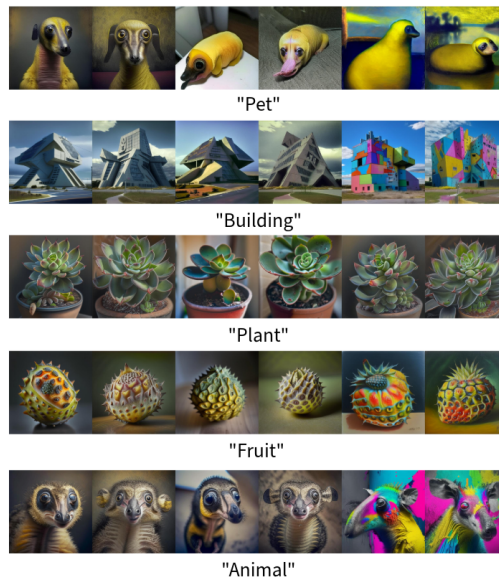


Figure 6: Set of Images Generated for Different Concepts using CreativeLab

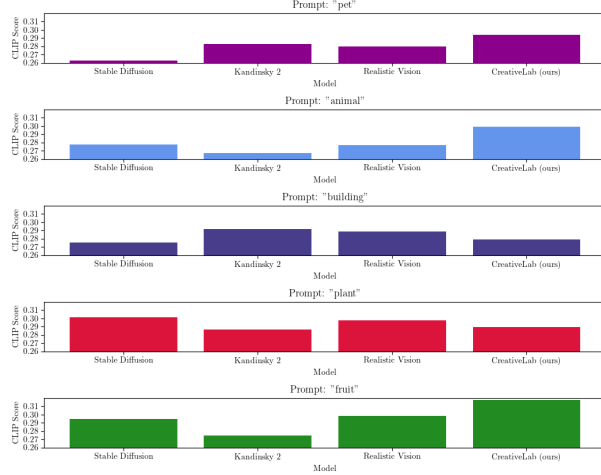


Figure 7: Positive CLIP Similarity Scores per Prompt

4.5 Implementation Insights and Extended Testing

Implementation involved restructuring the ConceptLab framework to integrate centroid-based clustering and prompt selection strategies. This included extensive testing of different embedding configurations (CLIP vs. DINOv2) and loss function variants (cosine similarity vs. WGAN-inspired objective).

Ablation studies explored combinations of positive and negative prompts across both text and image modalities. The most effective setup involved using text embeddings for positive prompts and image cluster centroids for negative prompts. Grid search results confirmed that cosine similarity remained the most stable loss formulation for both convergence and output quality.

Prompt throughput and training duration were also examined. Figures 8 and 9 show that training for only 300 steps often led to underdeveloped features, while longer training (e.g., 2,000 steps) could introduce noise and spurious textual artifacts. Optimal configurations balanced expressiveness and coherence.

5 Conclusions and Future Work

In this paper, we present a novel approach to concept generation by combining the ideas in [4] and [1]. Adopting the clustering approach in [4] allows us to find an image that is most representative of its class, while using a similar workflow in [1] allows us to generate unique images of a generalized concept. By using a positive text prompt and negative image prompts during image generation, we are able to develop new and creative images of various concepts.

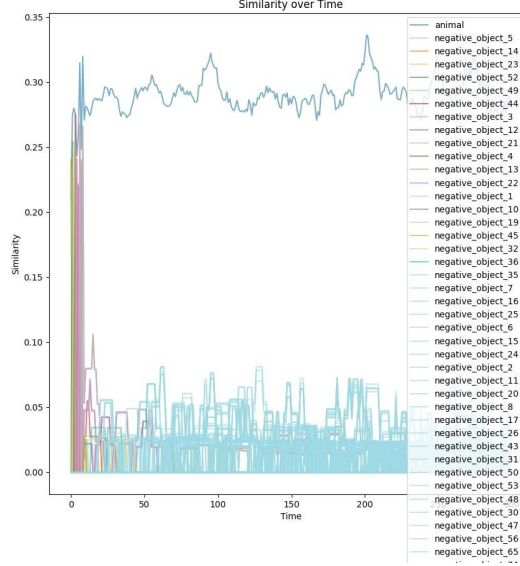
It is noticeable that the preexisting methods of creative image generation require a specific prompting to make advancements towards the final output. This was a particular theme in [1]. These methodologies are necessarily supervised in this manner, and assume that a label is inherent to preexisting concepts we do not want to generate in our final image. Our methodology rejects this idea, allowing for flexibility in generating images. CreativeLab was able to produce similarly creative imagery for similar concepts (like pet, animal, and fruit, for instance) to preexisting methodologies, while being unsupervised and not using specific prompting.

Future areas of improvement for our model include fine-tuning the hyperparameters further, conducting a user study to quantitatively verify the creativity of the generated images, reducing spurious results, and running a more rigorous statistical analysis to better quantify how significant our results are.

Certainly, from producing storyboards for films to children’s book illustrations, creative image generation has a myriad of applications in the entertainment and education sectors. Therefore, we hope our work can aid in future research in those areas and inspire new ideas in image synthesis.



(a) Images of Animals Generated at 300 Steps



(b) Similarity over Time Graph at 300 Steps

Figure 8: Creativity vs. Training Steps (300)

Acknowledgements

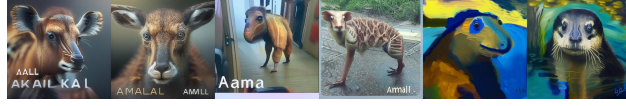
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One contributor, Ritika Allada, participated in early experimentation but chose not to be listed as an author in the final version. We respect this decision and acknowledge her early involvement.

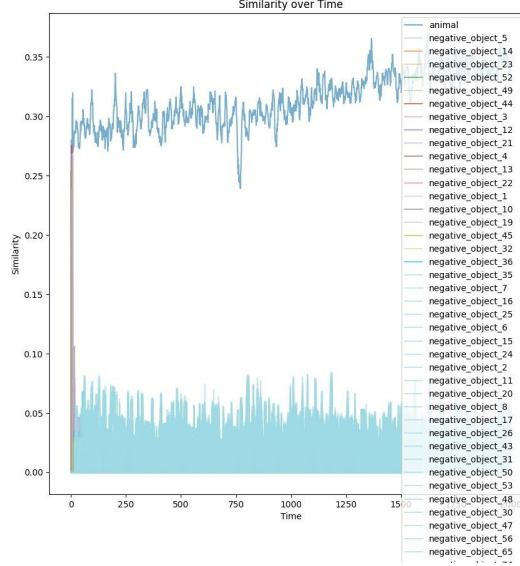
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(a) Images of Animals Generated at 2,000 Steps



(b) Similarity over Time Graph at 2,000 Steps

Figure 9: Creativity vs. Training Steps (2,000)

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